



Distributed Systems

Introduction

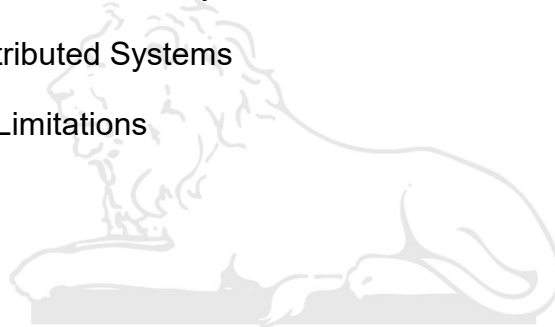
by Manoj Misra, IIT Roorkee



Recap



- What is a Distributed System
- Why Distributed Systems
- Issues / Limitations



Challenges



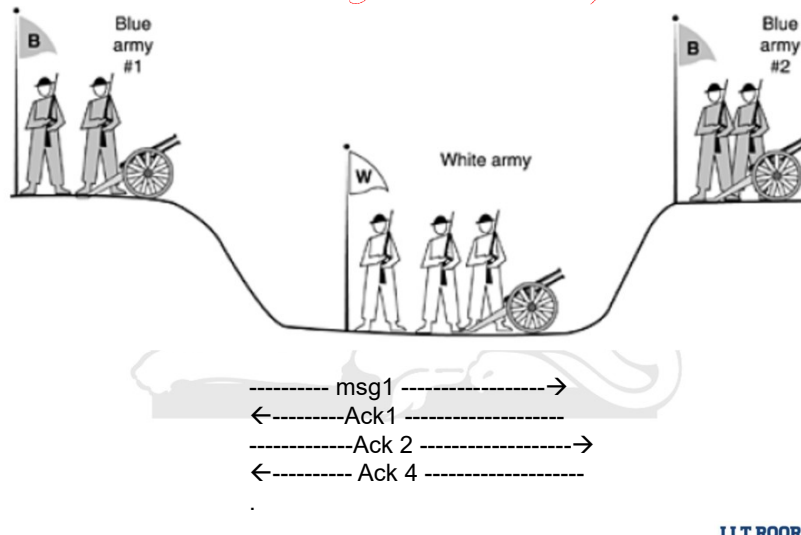
- Multiple computers
 - (No common physical clock, No shared memory)
 - ✓ Concurrent execution.
 - ✓ Independent failure.
 - ✓ Autonomous administration.
 - ✓ Heterogeneous.
 - ✓ Large numbers.
- Networked communication
 - ✓ Asynchronous
 - ✓ Unreliable
 - ✓ Insecure
- Common goal
 - ✓ Consistency
 - ✓ Transparency

IIT ROORKEE ■ ■ ■

Challenges: The two-army problem

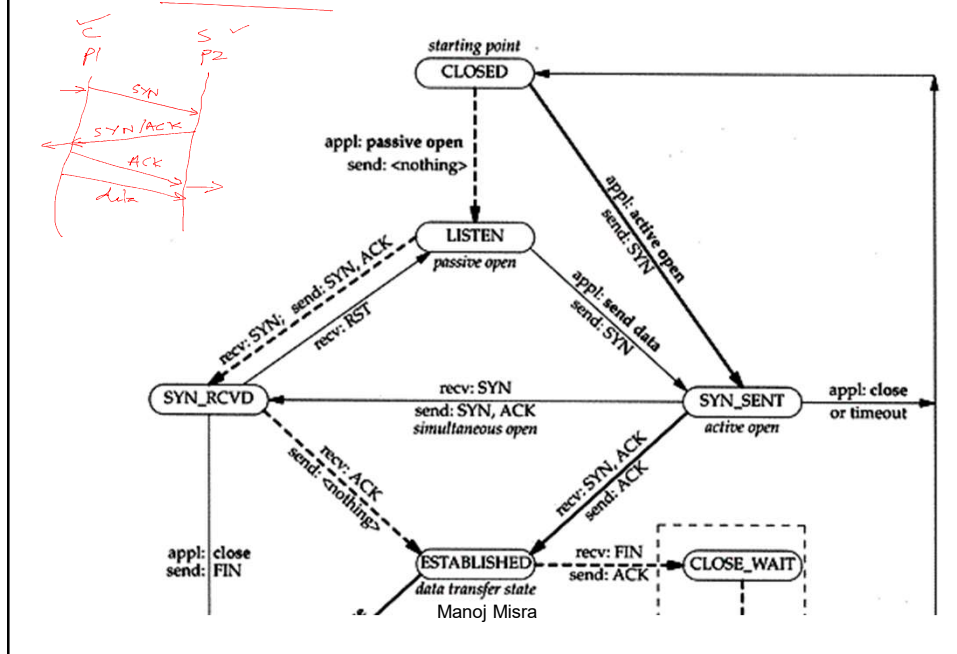


The two generals' prob./paradox



IIT ROORKEE ■ ■ ■

- The two-army problem



Challenges: Distributed Search



- A set of elements stored on a single machine
- A set of elements distributed and stored across multiple machines
- Need to search X
- How to know about the set of machines broadcast
- How to access them A, B, C, D directory
- How to get the information
- Updating X?
- X is replicated

Transparency in a Distributed System [5]



Transparency	Description
Access	Hide differences in data representation and how a resource is accessed
Location	Hide where a resource is located — Common name space
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
Replication	Hide that a resource is replicated
Concurrency	Hide that a resource may be shared by several competitive users
Failure	Hide the failure and recovery of a resource

ISO 1995

RPC $\frac{REQ}{BC}$ ✓ Socket API

- Trade-off between degree of transparency and performance of a system
- Transparency is preferable but may not be desirable always¹
- The transparencies could be defined and implemented according to the distributed system under development²